CAN7032/7132 Computational genomics, transcriptomics and evolution, 2025/2026
Coursework assignment

Submission deadline: 4pm of Thursday 26th Feb 2026 (UK time)

<u>This coursework assignment consists of 2 short answer questions (SAQs) and 4 long answer questions (LAQs). SAQs are to test your knowledge and understanding of the topics covered by this module, and LAQs are to assess your analytic skill and abilities.</u>

SAQ 1: Briefly describe the characteristics of the cancer genome (10 marks)

Cancer develops gradually as cells accumulate mutations over time. If a mutation gives a cell a growth or survival advantage, that cell is more likely to expand, so the tumour evolves through a process similar to natural selection.

Cancer genomes mutations occur at multiple levels. At the smallest scale, there are single base substitutions (SBSs), in which one nucleotide is replaced by another. These changes are not entirely random. Different biological processes—such as exposure to DNA-damaging agents or defects in DNA repair pathways—leave characteristic mutational patterns in the genome. However, alterations in cancer extend far beyond individual base changes. Many tumors show large-scale structural rearrangements involving segments of chromosomes. Copy number alterations are also common, leading to the amplification of oncogenes or the deletion of tumor suppressor genes. In addition, whole chromosomes may be gained or lost, a condition known as aneuploidy. These large-scale disruptions contribute significantly to genomic instability. In some cases, when chromosomal instability becomes severe, cancer cells undergo whole-genome duplication (WGD). Although duplicating the entire genome may seem counterintuitive, this process can provide a buffering effect. By doubling their genetic material, cancer cells may better tolerate subsequent chromosomal losses and maintain essential functions despite ongoing instability. This adaptation highlights how dynamic and flexible the cancer genome can be. Importantly, cancer genomes are rarely uniform. Even within a single tumor, different cells can carry distinct sets of mutations. This genetic heterogeneity reflects continuous mutation and selection and allows tumors to adapt to environmental pressures, including therapy. As a result, cancer progression is not static but an ongoing evolutionary process.

SAQ 2:

1. **What is cancer immunotherapy? (2 marks)**

Cancer immunotherapy is a treatment that helps stimulate or restore the body's immune system so it can recognise and destroy tumour cells. Unlike chemotherapy, which directly targets rapidly dividing cells, immunotherapy works by boosting the immune response against cancer. It aims to overcome the immune suppression created by tumours and promote a stronger anti-tumour response.

2. **Give two examples of cancer immunotherapy, and also brief describe the underlying mechanism of each immunotherapy? (4 marks)**

Two examples of cancer immunotherapy are immune checkpoint inhibition and therapies targeting tumour-associated macrophages (TAMs).

The first example is immune checkpoint inhibition. In many tumours, CD8+ T cells can already be found in the tumour microenvironment, but their function is often suppressed. These cells normally have the ability to kill tumour cells and are linked to better prognosis when they are present in high numbers. However, inhibitory signals in the tumour microenvironment can limit their activation. Immune checkpoint inhibitors block these signals, allowing cytotoxic T cells to regain their function and improve anti-tumour immune responses. This type of treatment is especially effective in tumours with an inflamed phenotype, where T cells are already present in the tumour tissue.

The second example is therapies targeting tumour-associated macrophages. TAMs are common in many cancers and are often linked to poor prognosis because they promote tumour growth, angiogenesis, and immune suppression. They can also inhibit effective T-cell responses and support tumour progression. For this reason, some therapeutic approaches aim to reduce the number of TAMs or change them into a more anti-tumour phenotype, which can help strengthen overall anti-tumour immunity.

3. **Briefly describe strategies that tumour cells take to evade immune surveillance. (4 marks)**

Tumour cells can avoid immune surveillance in several ways. One important way is creating an immunosuppressive tumour microenvironment. They recruit regulatory T cells and myeloid-derived suppressor cells, which reduce the activity of cytotoxic T cells and weaken the the immune response against the tumour. Dendritic cells in the tumour can also become defective, so they are less effective at activating T cells. Sometimes T cells are present, but they cannot enter the tumour core. Changes in the extracellular matrix and stromal structure can block their access and result in an immune-excluded phenotype. Tumours may also induce immune tolerance in lymph nodes and form abnormal blood vessels, making it more difficult for immune cells to infiltrate the tumour and function efficiently.

LAQ 1:

High grade serous ovarian cancer cell lines Kuramochi and Snu119 were single cell sorted, such that one individual cell was dispensed into one well of a 96 well plate to form a new clonal population (a 'clone'). Each new clone was expanded into a new clonal population over 22 cell division cycles. Single cell DNA sequencing (low pass whole genome sequencing) was performed on small subsets of (i) the parental population that the single cell was derived from, and (ii) the new clonal populations (C8 & E10 (Kuramochi) and A12 & F12 (Snu119)). (**Total 25 marks)**

Get the data required for this question from '`/data/teaching/bci_teaching/Computational_genomics/`'. You can find further instructions on how to transfer the data to your hpc folder in the slide no 8 in 'Practical 2_Study Tumour Evolution using single cell sequencing_2026.pptx' file.

1. **5 marks** are available for successfully running the data through Aneufinder and MEDICC2 and selecting the relevant data to contribute to the following questions.

```
library(AneuFinder)

library(BSgenome.Hsapiens.UCSC.hg38)

Aneufinder(inputfolder = "Kuramochi_E10/",

 outputfolder = "Kuramochi_E10/Aneufinder_output/",

 assembly = 'hg38',

 numCPU = 2,

 binsizes = 1e6,

 blacklist = "hg38-blacklist.v2.bed",

 correction.method = 'GC',

 GC.BSgenome = BSgenome.Hsapiens.UCSC.hg38,

 refine.breakpoints = FALSE,

 method = 'edivisive',

 chromosomes =
c("1","2","3","4","5","6","7","8","9","10","11","12","13","14","15","16","17","18","19","20","2
1","22","X"))

Aneufinder(inputfolder = "Kuramochi_C8/",

 outputfolder = "Kuramochi_C8/Aneufinder_output/",

 assembly = 'hg38',

 numCPU = 2,

 binsizes = 1e6,
```

```
  blacklist = "hg38-blacklist.v2.bed",
  correction.method = 'GC',
  GC.BSgenome = BSgenome.Hsapiens.UCSC.hg38,
  refine.breakpoints = FALSE,
  method = 'edivisive',
  chromosomes =
c(“1","2","3","4","5","6","7","8","9","10","11","12","13","14","15","16","17","18","19","20","2
1","22","X"))
Aneufinder(inputfolder = "Snu119_A12/",
  outputfolder = "Snu119_A12/Aneufinder_output/",
  assembly = 'hg38',
  numCPU = 2,
  binsizes = 1e6,
  blacklist = "hg38-blacklist.v2.bed",
  correction.method = 'GC',
  GC.BSgenome = BSgenome.Hsapiens.UCSC.hg38,
  refine.breakpoints = FALSE,
  method = 'edivisive',
  chromosomes =
c(“1","2","3","4","5","6","7","8","9","10","11","12","13","14","15","16","17","18","19","20","2
1","22","X"))
Aneufinder(inputfolder = "Snu119_F12/",
  outputfolder = "Snu119_F12/Aneufinder_output/",
  assembly = 'hg38',
  numCPU = 2,
  binsizes = 1e6,
  blacklist = "hg38-blacklist.v2.bed",
  correction.method = 'GC',
  GC.BSgenome = BSgenome.Hsapiens.UCSC.hg38,
  refine.breakpoints = FALSE,
  method = 'edivisive',
```

```
chromosomes =
c("1","2","3","4","5","6","7","8","9","10","11","12","13","14","15","16","17","18","19","20","21
","22","X"))
```

2. Which type of alterations can you identify in the single cell whole genome sequencing data that is output from AneuFinder from the ovarian cancer genomes? Which types of alteration(s) **cannot** be identified in these profiles, and why? **(4 marks)**

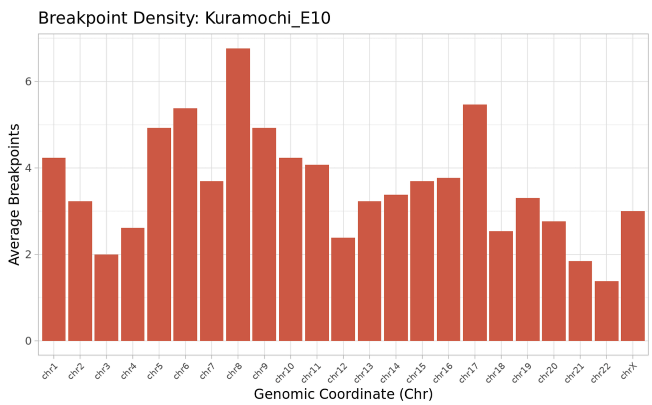
```
f_E10 = list.files('Kuramochi_E10/Aneufinder_output/MODELS/method-edivisive/',
full.names=TRUE)

load_E10 = loadFromFiles(f_E10)

plot(load_E10[[1]], type='profile')

f_C8 = list.files('Kuramochi_C8/Aneufinder_output/MODELS/method-edivisive/',
full.names=TRUE)

load_C8 = loadFromFiles(f_C8)

plot(load_C8[[1]], type='profile')

f_A12 = list.files('Snu119_A12/Aneufinder_output/MODELS/method-edivisive/',
full.names=TRUE)

load_A12 = loadFromFiles(f_A12)

plot(load_A12[[1]], type='profile')

f_F12 = list.files('Snu119_F12/Aneufinder_output/MODELS/method-edivisive/',
full.names=TRUE)

load_F12 = loadFromFiles(f_F12)

plot(load_F12[[1]], type='profile')
```

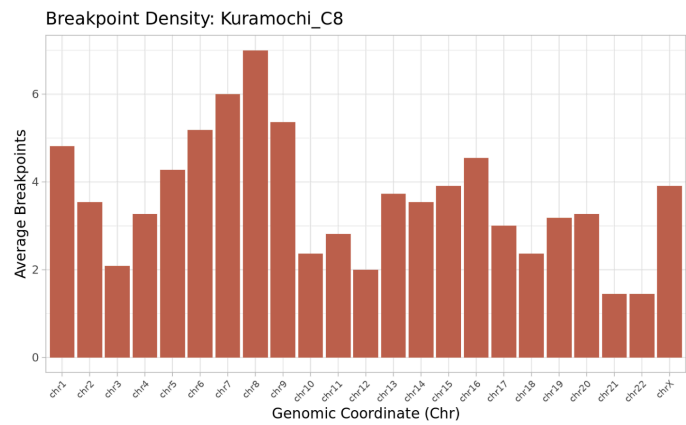
Single-cell whole genome sequencing data analysed with AneuFinder can identify whole-chromosome aneuploidies, large-scale segmental gains and losses (CNAs), and the precise locations of genomic breakpoints. However, point mutations (SNVs) cannot be detected because the sequencing depth is too shallow to call individual base-pair changes. Balanced rearrangements, such as inversions and translocations, are also not identifiable because they do not change the total amount of DNA; AneuFinder detects alterations based on read depth (DNA quantity), which remains constant in these cases. Small indels cannot be detected either, as they are smaller than the megabase-scale analysis bin size and are therefore filtered out as background noise.

3. Calculate the number of genomic breakpoints (location where copy number change is detected) per cell and per chromosome from the AneuFinder browser CNV files. Are the CNAs distributed evenly across the chromosomes? Demonstrate this using representative figures or graphs. (**5 marks**)

Kuramochi E10) Avg_Chrom: 3.602007, Avg_Cell: 82.84615

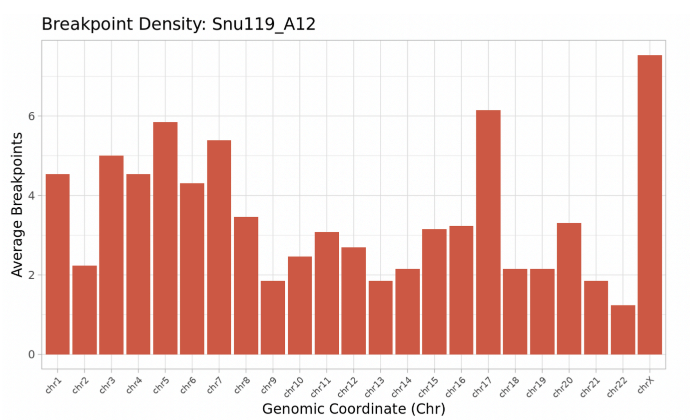
Kuramochi C8) Avg_Chrom: 3.612648, Avg_Cell: 83.09091

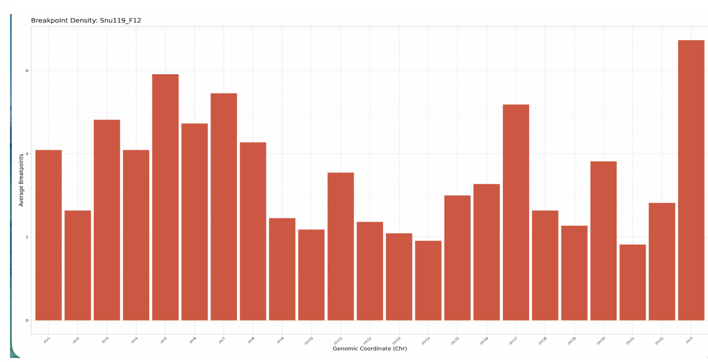




Snu119 A12) Avg_Chrom: 3.48495, Avg_Cell: 80.15385

Snu119 F12) Avg_Chrom: 3.56917, Avg_Cell: 82.09091





The number of genomic breakpoints per chromosome and per cell was calculated from the AneuFinder CNV files for each clone.

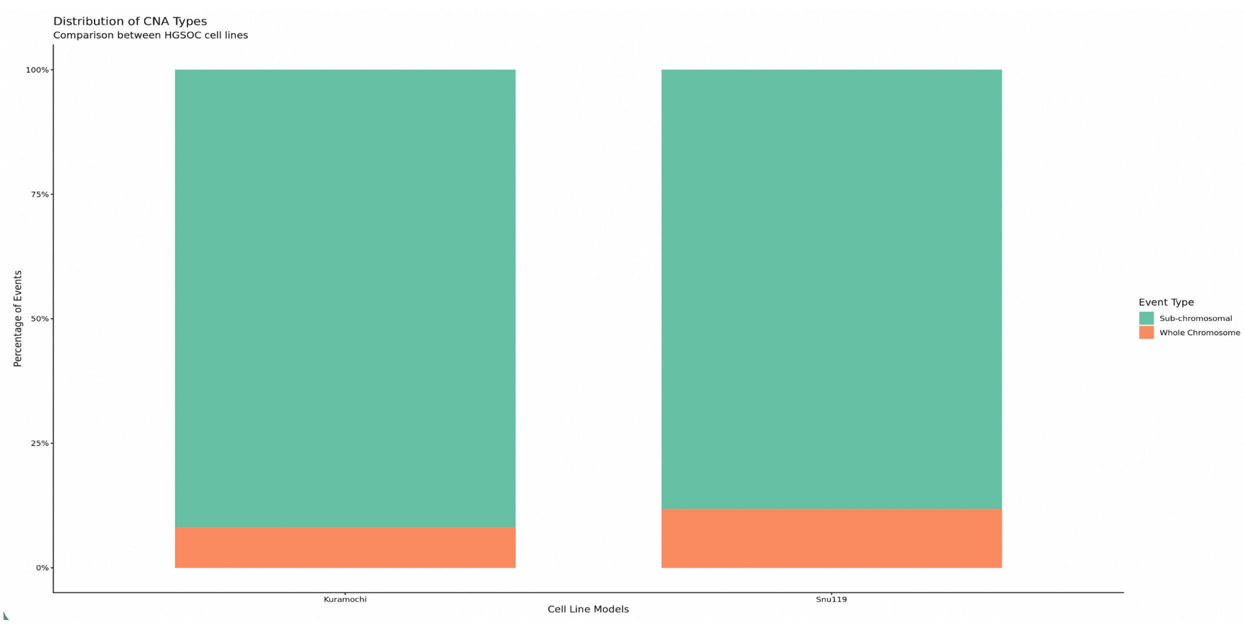
On average, the number of breakpoints per chromosome ranges from 3.48 to 3.61 (Kuramochi E10: 3.602007; Kuramochi C8: 3.612648; Snu119 A12: 3.48495; Snu119 F12: 3.56917).

Per cell (whole genome), the mean number of breakpoints ranges from 80.15385 (Snu119 A12) to 83.09091 (Kuramochi C8), with Kuramochi E10 at 82.84615 and Snu119 F12 at 82.09091.

CNAs are not evenly distributed across chromosomes. The bar charts show that some chromosomes, such as chromosome 8 in Kuramochi, have higher breakpoint frequencies than others. If the distribution were even, all bars would be similar in height, but the variation indicates genomic instability hotspots in these cell lines.

4. What is the size range of CNAs across the two HGSOC cell lines and clones and how often are these whole chromosome alterations vs sub-chromosomal? Demonstrate this using representative figures or graphs. (**4 marks**).

The CNA size range in the two HGSOC cell lines varies from approximately 2 Mb to 147 Mb in Kuramochi and from 2 Mb to 157 Mb in Snu119.

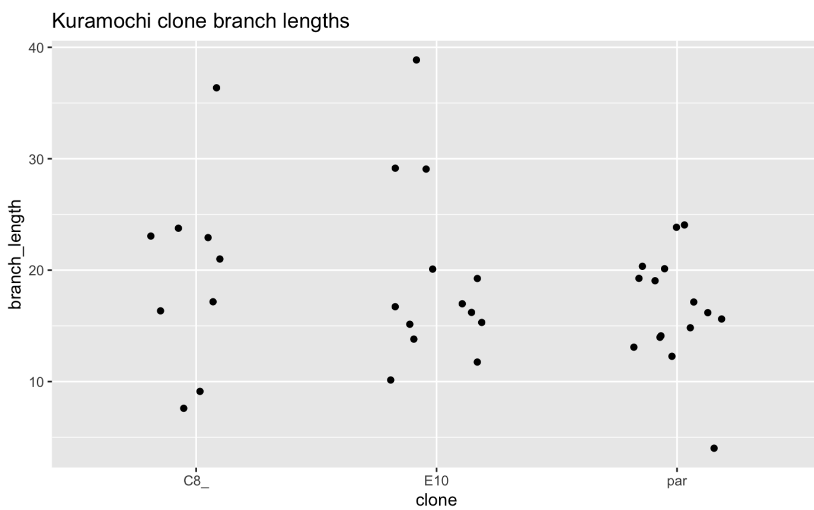


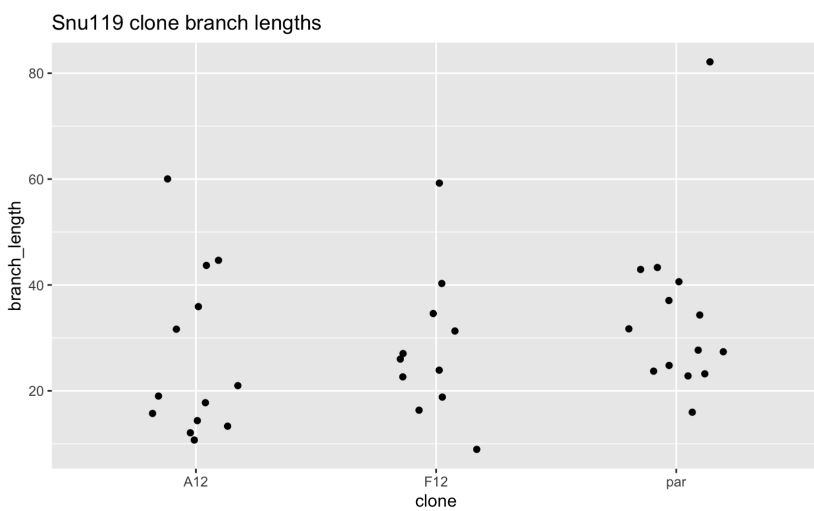
The lower limit (~2 Mb) shows that many alterations are focal and sub-chromosomal. The upper limit (~150 Mb) is close to the size of whole chromosomes, indicating that whole-chromosome gains or losses do occur. However, based on the CNA segmentation analysis, most alterations in both cell lines are sub-chromosomal rather than whole-chromosome events. Whole-chromosome alterations represent only a small fraction of the total CNAs.

Overall, this suggests that genomic instability in these HGSOC cell lines is mainly driven by chromosomal fragmentation rather than simple whole-chromosome aneuploidy.

5. Using metrics produced by MEDICC2, calculate the total branch lengths per cell from the branch_lengths.tsv (be careful to remove first internal branch which indicates the length from diploid to the ancestral cancer cell). Produce scatter plots for the clone populations of Kuramochi & Snu119. Comment on the significance of the branch lengths of the populations and how this could be informative for inferring rates of genomic alteration, given that all clones were grown for 22 cell divisions (**7 marks).**

The mean total branch lengths were higher in the Snu119 clones (A12: 26.2, F12: 28.1) compared to the Kuramochi clones (C8: 19.7, E10: 19.4). The scatter plots show a clear shift towards longer branch lengths in Snu119 relative to Kuramochi.





Since all clones were grown for 22 cell divisions, the longer branch lengths in Snu119 suggest that this cell line accumulates copy number alterations at a higher rate per division. In contrast, the similar and lower branch lengths observed in Kuramochi indicate a more stable genomic background. Overall, these differences suggest that Snu119 has a higher rate of genomic instability compared to Kuramochi under the same growth conditions.

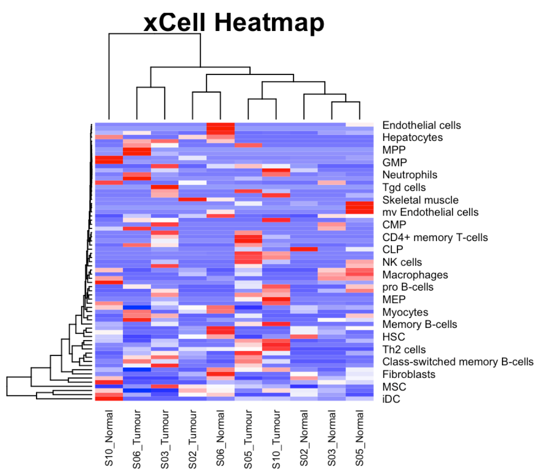
LAQ 2:
In the CAN7031 “omics analytics” coursework assignment, you were given the raw count matrix of 10 matched normal and tumour samples from 5 patients, S02, S03, S05, S06 and S10. Using the normalised gene expression matrix you created in the task (e.g., RPKM or CPM log2 transformed or not transformed) to perform the bulk RNA deconvolution analysis. You are required to answer the following questions. (**Total 20 marks)**

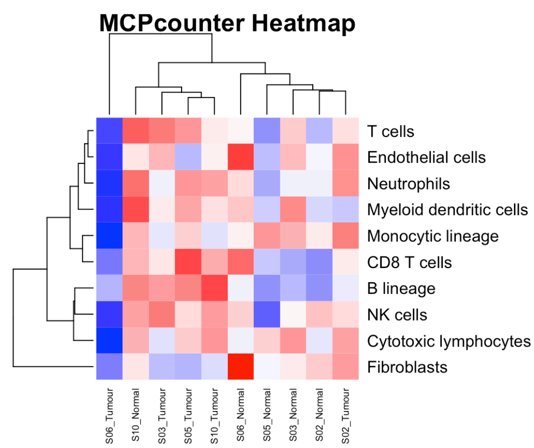
(1) Use two individual and one consensus methods to perform the bulk RNA deconvolution. And produce heatmaps based on the cellular estimates, with samples clustered based on the profile. Briefly describe the patterns you can see. (Hint: pay attention to the input format depending on the methods you use, e.g., logged or unlogged values) **(8 marks)**

Two individual deconvolution methods (xCell, MCPcounter) and one consensus method (Decosus) were applied to estimate cellular composition. Log2-transformed normalized counts were used for xCell and MCPcounter, whereas unlogged normalized counts were used for Decosus as required by the method.

```
library(xCell)

xcell_e <- xCellAnalysis(norm_logged)

dim(xcell_e)

heatmap(xcell_e, col=colorRampPalette(c("blue","white","red"))(100), main="xCell Cellular Estimates")

library(MCPcounter)

mcp_e <- MCPcounter.estimate(norm_logged,featuresType="HUGO_symbols")

dim(mcp_e)

heatmap(as.matrix(mcp_e),col=colorRampPalette(c(“blue","white","red"))(100),main="MCPcounter Cellular Estimates")

library(Decosus)

exp_data_unlog1 <- as.data.frame(norm_unlogged)

exp_data_unlog1$Gene <- rownames(exp_data_unlog1)

exp_data_unlog1 <- exp_data_unlog1[, c("Gene",colnames(norm_unlogged))]

est_Decosus <- Decosus::cosDeco(x = exp_data_unlog1,rnaseq = T,plot = TRUE)
```

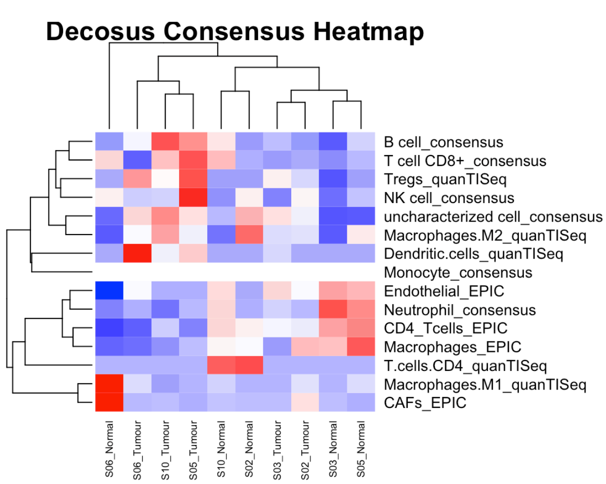
```
est_Decosus_cells <- est_Decosus$main_cells

est_Decosus_samples <- est_Decosus$main_samples

est_Decosus_cells <- as.matrix(est_Decosus_cells)

est_Decosus_cells[is.nan(est_Decosus_cells)] <- NA

est_Decosus_cells[is.na(est_Decosus_cells)] <- 0
```





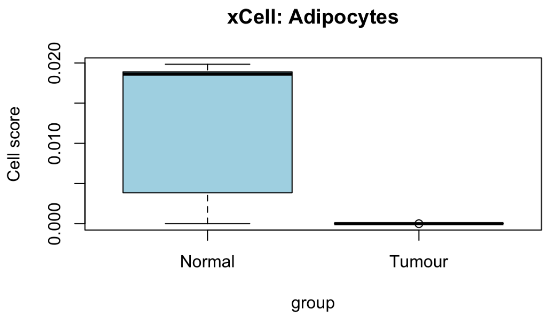

```
heatmap(as.matrix(est_Decosus_cells),col=colorRampPalette(c("blue","white","red"))(100),main
= "Decosus Consensus Estimates")
```

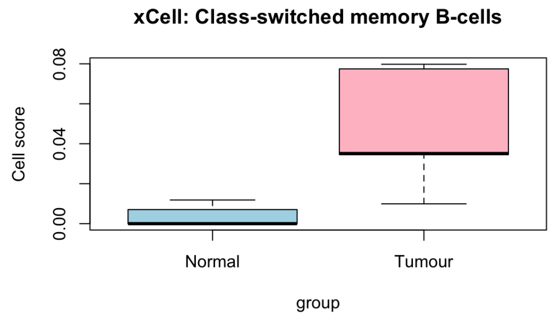
Heatmaps based on the cellular estimates showed that samples clustered according to their cellular composition. Across all methods, tumour samples generally grouped together and showed higher immune-related signatures compared to normal tissues. In particular, MCPcounter indicated increased levels of T cells, CD8 T cells, NK cells, and cytotoxic lymphocytes in tumour samples.

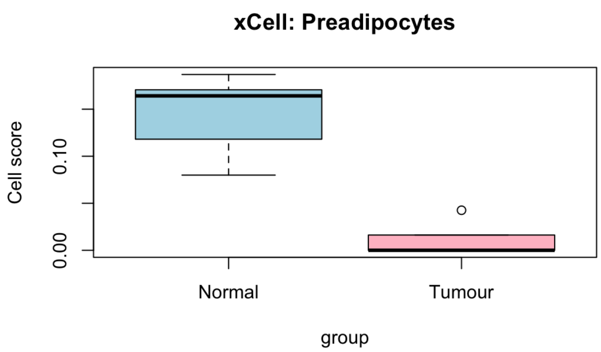


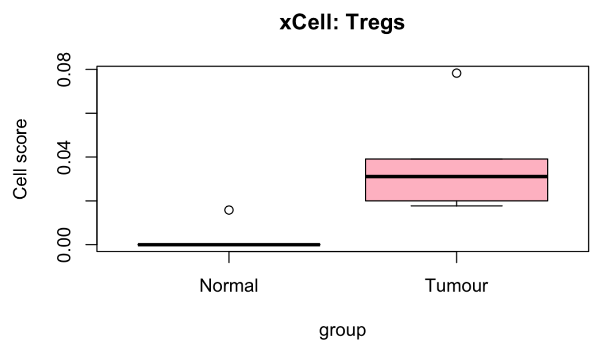
The Decosus consensus results showed more consistent clustering patterns, which suggests that the tumour-associated immune infiltration observed is robust and reliable.

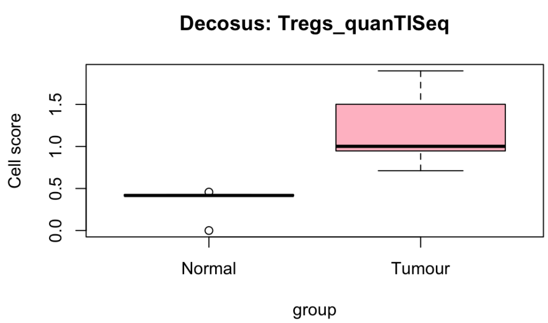
(2) Perform student t-test on the cellular estimates between tumour and normal groups for these estimates from the three methods above. List the cell types with the significant differences ($p < 0.05$), and produce boxplots for these significant ones. **(5 marks)**

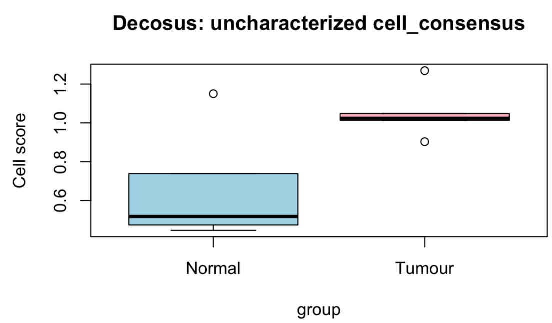












Student's t-test was performed to compare cellular estimates between tumour and normal samples. In the xCell analysis, four cell types showed significant differences ($p < 0.05$): Adipocytes, Class-switched memory B-cells, Preadipocytes, and Tregs.
In the Decosus consensus analysis, two cell types were significantly different between groups: uncharacterized cell_consensus and Tregs_quanTISeq.
In contrast, no cell types reached statistical significance in the MCPcounter analysis at the $p < 0.05$ threshold.

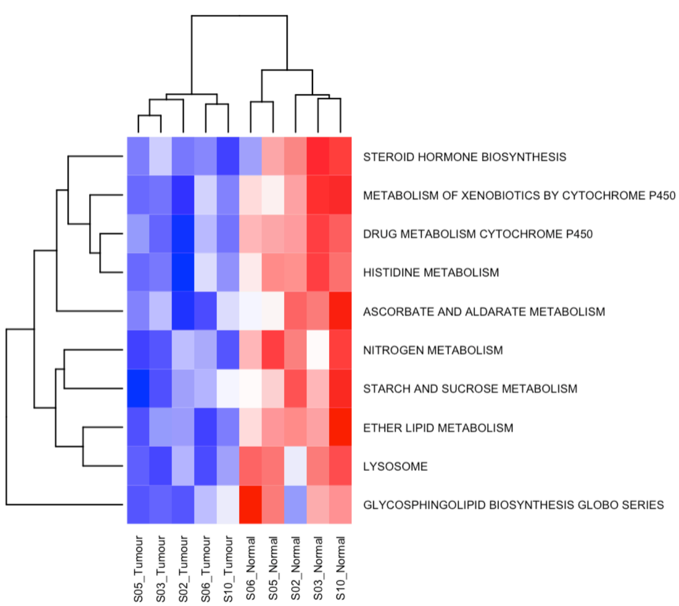
(3) Perform the GSVA analysis (i.e., tumour vs. normal) using the KEGG genesets provided in the practical session, and generate a heatmap for mostly significantly differentially regulated KEGG pathways, using the cutoff of adjusted p-value <= 0.01. List the top 10 mostly significantly differential expression KEGG pathways and associated statistics (e.g., fold change, p-value, adjusted p-value etc) in a table. **(7 marks)**

GSVA was performed using the KEGG gene sets provided in the practical session to compare tumour and normal samples. Pathways with an adjusted p-value ≤ 0.01 were considered statistically significant.

Description: df [10 × 6]

	logFC <dbl>	AveExpr <dbl>	t <dbl>	P.Value <dbl>	adj.P.Val <dbl>	B <dbl>
KEGG_DRUG_METABOLISM_CYTOCHROME_P450	-0.5469246	0.019638708	-5.385539	0.0002850735	0.02195747	0.6939958
KEGG_NITROGEN_METABOLISM	-0.5874675	0.027079033	-5.263280	0.0003404135	0.02195747	0.5378852
KEGG_ASCORBATE_AND_ALDARATE_METABOLISM	-0.5800881	0.068179014	-5.236213	0.0003541527	0.02195747	0.5029695
KEGG_STARCH_AND_SUCROSE_METABOLISM	-0.4880543	0.023440381	-4.782958	0.0006977582	0.02689454	-0.1006452
KEGG_HISTIDINE_METABOLISM	-0.5033017	-0.012288294	-4.618039	0.0008995539	0.02689454	-0.3290395
KEGG_ETHER_LIPID_METABOLISM	-0.4300213	0.020897019	-4.596713	0.0009298543	0.02689454	-0.3589075
KEGG_METABOLISM_OF_XENOBIOTICS_BY_CYTOCHROME_P450	-0.4872259	0.031624516	-4.434184	0.0011994390	0.02689454	-0.5890161
KEGG_LYSOSOME	-0.4447488	-0.001065876	-4.332640	0.0014088419	0.02689454	-0.7349617
KEGG_STEROID_HORMONE_BIOSYNTHESIS	-0.5184088	0.047952127	-4.306200	0.0014694630	0.02689454	-0.7732311
KEGG_GLYCOSPHINGOLIPID_BIOSYNTHESIS_GLOBO_SERIES	-0.4759416	-0.011791047	-4.256786	0.0015902446	0.02689454	-0.8450484

No KEGG pathways met this strict adjusted p-value threshold. Therefore, the top 10 pathways ranked by adjusted p-value were selected for exploratory analysis. These pathways were mainly associated with metabolic processes, including drug metabolism, nitrogen metabolism, carbohydrate metabolism, lipid metabolism, and steroid hormone biosynthesis. Most of these pathways showed negative log fold change values, indicating reduced pathway enrichment in tumour samples compared to normal tissues.



The heatmap of the top-ranked pathways showed clear differences in enrichment patterns between tumour and normal samples, although these differences did not reach the adjusted p-value ≤ 0.01 significance threshold.

LAQ 3:
In the data folder under the "Assessments" tab, you will find the VCF files of four skin tumour samples, namely WD06, WD13, WD14, WD22, with somatic mutations called against the matched normal reported with the coordinates **in the hg38 format**.

Perform the analysis of tumour neoantigen analysis using MuPeXi, selecting two major HLA-A alleles, HLA-A02:01,HLA-A24:02, and peptide length of 8-11 mers. No need to provide the expression file. (**Total 18 marks)**

(1) How many tumour neoantigens can you find for each tumour? How many strong tumour neoantigens are there for each tumour? (Hint: use the rank score cutoff values mentioned in the lecture/practical). **(7 marks)**

After filtering the dataset using the specified rank score cut-offs, tumour neoantigens were defined as peptides with Norm_MHCrank_EL > 2 and Mut_MHCrank_EL < 2, indicating stronger predicted binding of the mutant peptide compared to the normal peptide. Using these criteria, 561 predicted tumour neoantigens were identified in WD06, 360 in WD13, 64 in WD14, and 483 in WD22. To identify strong tumour neoantigens, a stricter threshold of Mut_MHCrank_EL < 0.5 was applied. Based on this cut-off, 66 strong neoantigens were found in WD06, 37 in WD13, 8 in WD14, and 70 in WD22.

(2) For all the strong tumour neoantigens identified, what are the affected genes? What are the neoantigen peptides? What are the rank scores in tumour and normal, respective, for these strong tumour neoantigens? Create a table for all the strong tumour neoantigens, with all the essential statistics, e.g., including rank scores, and with one column indicating the tumour sample. **(7 marks)**

All strong tumour neoantigens were compiled into a table named StrongTumorNeoantigens_Stats.csv. The table includes the affected gene, the neoantigen peptide sequence, the tumour rank score (Mut_MHCrank_EL), the normal rank score (Norm_MHCrank_EL), and a column indicating the tumour sample (WD06, WD13, WD14, or WD22).This table summarises all strong tumour neoantigens with the key statistics and allows direct comparison between tumour samples.

(3) Are there any shared tumour neoantigens amongst the four tumour samples? What are the potential implications if neoantigen-based therapies are designed? **(4 marks)**

No shared tumour neoantigens were found among the four tumour samples (WD06, WD13, WD14 and WD22). Each tumour has its own unique set of neoantigens.

This suggests that neoantigen-based therapies would likely need to be personalised. Because the neoantigens differ between tumours, designing a single universal vaccine would be challenging. Instead, patient-specific immunotherapy approaches targeting the unique neoantigens in each tumour would probably be more effective.

LAQ 4:

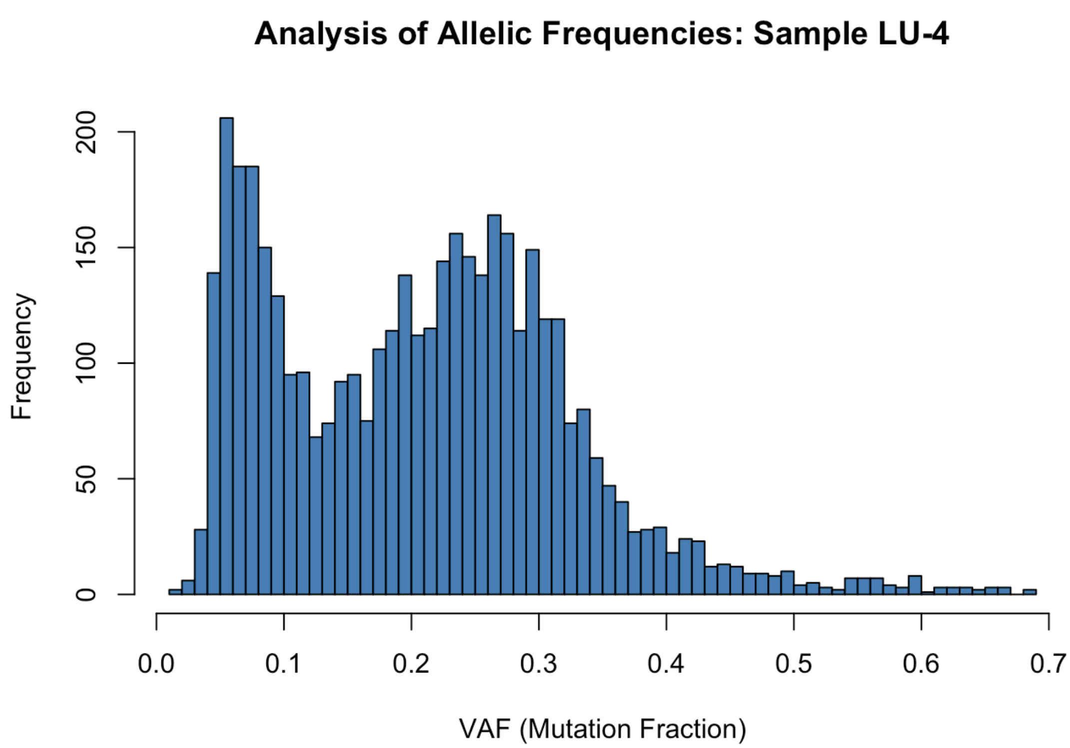
MClust tumour clonality anlaysis (**Total 17 marks**).

In the data folder under the "Assessments" tab, you will find "**LU-4.snv.txt**" that contains all somatic mutations identified for the LU-4 tumour sample, and "**mutations_SP4_LN1_LN2_together.txt**" that contains all mutations identified and shared betweel SP4 lymph node 1 (LN1) and lymph node 2 (LN2) tumour samples. This question is to investigate the clonal evolution patterns for a single sample "LU-4", and for comparing two sites, LN1 vs. LN2, of the SP4 patient, using the mClust analysis.

For the single sample analysis,

(1) Load the "LU-4.snv.txt" file, and run the mClust single sample clustering analysis using variant allele frequency (VAF) of all mutations identified. First, generate a histogram of all VAFs, and briefly comment on the VAF distribution (2 marks).

```
library(mclust)

lu4_data <- read.delim("LU-4.snv.txt", header=TRUE)

lu4_data$vaf <- lu4_data$t_alt_count / (lu4_data$t_alt_count + lu4_data$t_ref_count)

dim(lu4_data)
```



The VAF histogram shows a small group of mutations between 0.05 and 0.1. These are likely recent mutations found in a small group of tumour cells, suggesting that the tumour is still evolving. There is a wider peak around 0.25–0.3 VAF, which probably represents the main tumour clone. The fact that it is lower than the expected 0.5 could be due to low tumour purity or copy number alterations (CNAs). The wide shape of the peak may also mean that more than one subclone is present.

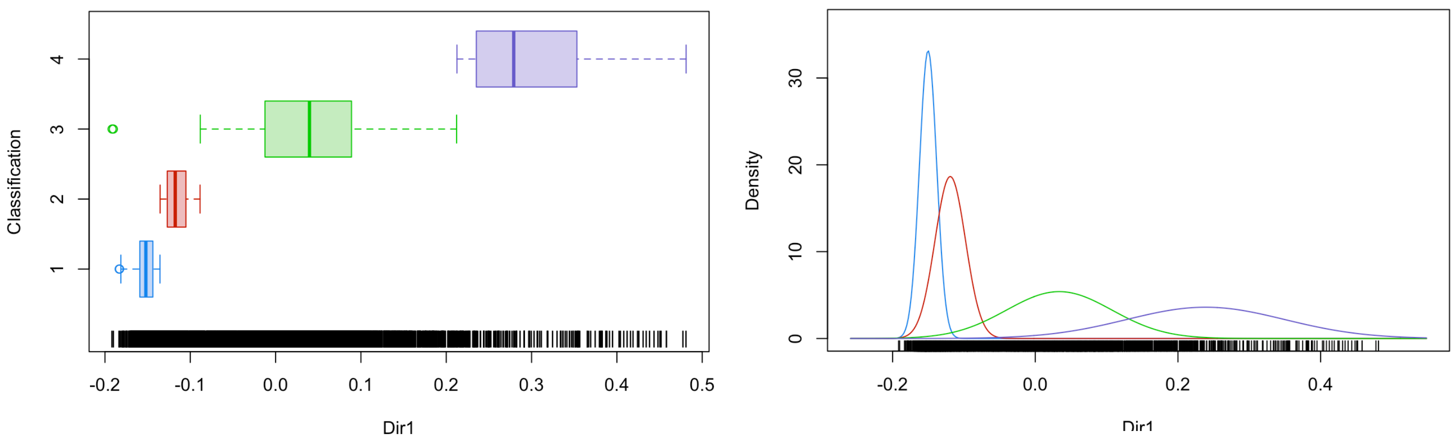
(2) Run the density estimation analysis via Gaussian finite mixture modelling, followed by dimension reduction clustering and classification analysis. How many clusters can you identify? What is the size (# of mutations) for each cluster? What is the mean VAF value for each cluster? Can you identify which are major clones (heterozygous or homozygous), which are subclones, and their associated cancer cell fractions (CCF)? You can summarise all these into a table as shown in the lecture/practical slides (**3 marks**).

```
mod_lu4 <- densityMclust(lu4_data$vaf)

plot(mod_lu4, what = "diagnostic", type = “cdf")

plot(mod_lu4, what = "diagnostic", type = “qq")
```

Two lines in the CDF plot are so close which means that the model is very accurate. Points and the line (QQ plot) are aligned well that means a good fit

```
mod_lu4dr <- MclustDR(mod_lu4)

summary(mod_lu4dr)

cluster_means <- mod_lu4dr$mu

print(cluster_means)

plot(mod_lu4dr, what = “classification")

plot(mod_lu4dr, what = “density")

summary_table <- data.frame(

  Cluster = c("Cluster 1", "Cluster 2", "Cluster 3", "Cluster 4"),

  Size = c(624, 595, 2823, 165),

  VAF = c(0.058, 0.089, 0.242, 0.447))
```

```
trunk_vaf <- max(summary_table$VAF)
summary_table$ScaledVAF<- summary_table$VAF / trunk_vaf
summary_table$CCF <- summary_table$ScaledVAF* 100
summary_table$Info <- c("Subclone", "Subclone","Dominant subclone","Trunk")
summary_table$Zygosity <- "Heterozygous"
print(summary_table)
```

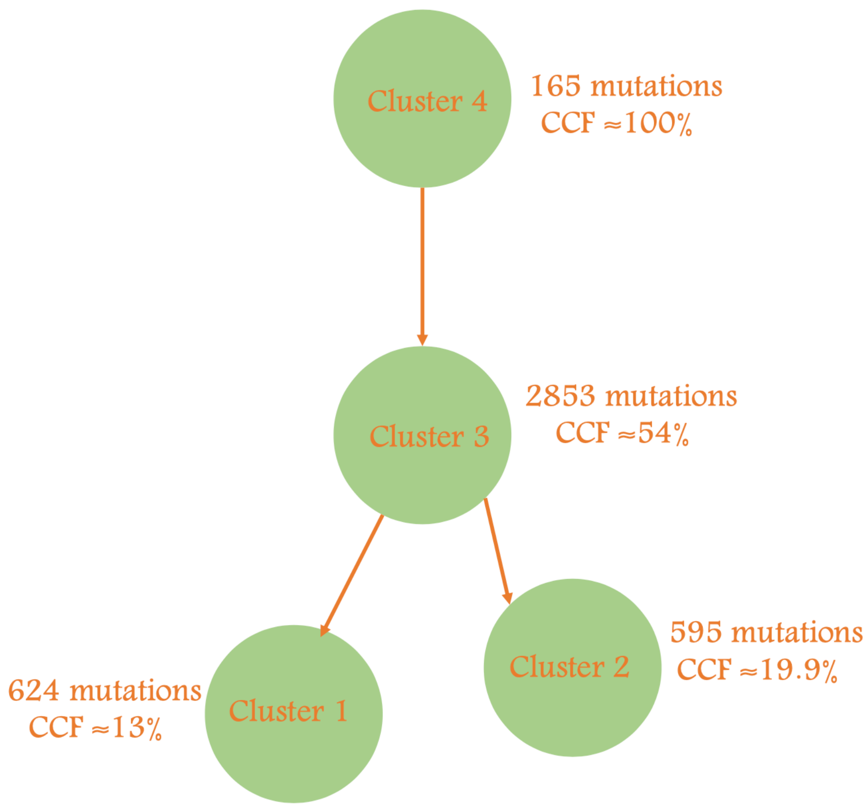



```
##     Cluster Size   VAF ScaledVAF       CCF              Info     Zygosity
## 1 Cluster 1  624 0.058 0.1297539  12.97539          Subclone Heterozygous
## 2 Cluster 2  595 0.089 0.1991051  19.91051          Subclone Heterozygous
## 3 Cluster 3 2823 0.242 0.5413870  54.13870 Dominant subclone Heterozygous
## 4 Cluster 4  165 0.447 1.0000000 100.00000             Trunk Heterozygous
```

There are four clusters. All four clusters are heterozygous. Cluster 4 (mean VAF ~0.45) represents the main clonal population (clonal trunk), while Clusters 1–3 have lower mean VAF values and represent subclones. As no copy number data were available, a diploid state was assumed, which fits the mclust distribution. Under this assumption, cancer cell fraction (CCF) can be estimated directly from the scaled VAF values. If scaling is not applied, VAF would need to be multiplied by 2 to approximate CCF, although this is less accurate.

(3) Finally, please draw a tumour evolutionary tree to illustrate the clustering patterns listed above. Within the tree, specifiy clearly the cluster names/numbers and asscoated CCF values (**2 marks**).

Cluster	size	Mean Vaf	Scaled Vaf	CCF %	Info	Zygosity
4	165	0.447	1.0	100	Trunk	Heterozygous
3	2823	0.242	0.54	54.1	Subclone(Major	Heterozygous
2	595	0.089	0.19	19.9	Subclone	Heterozygous
1	624	0.058	0.12	12.9	Subclone	Heterozygous

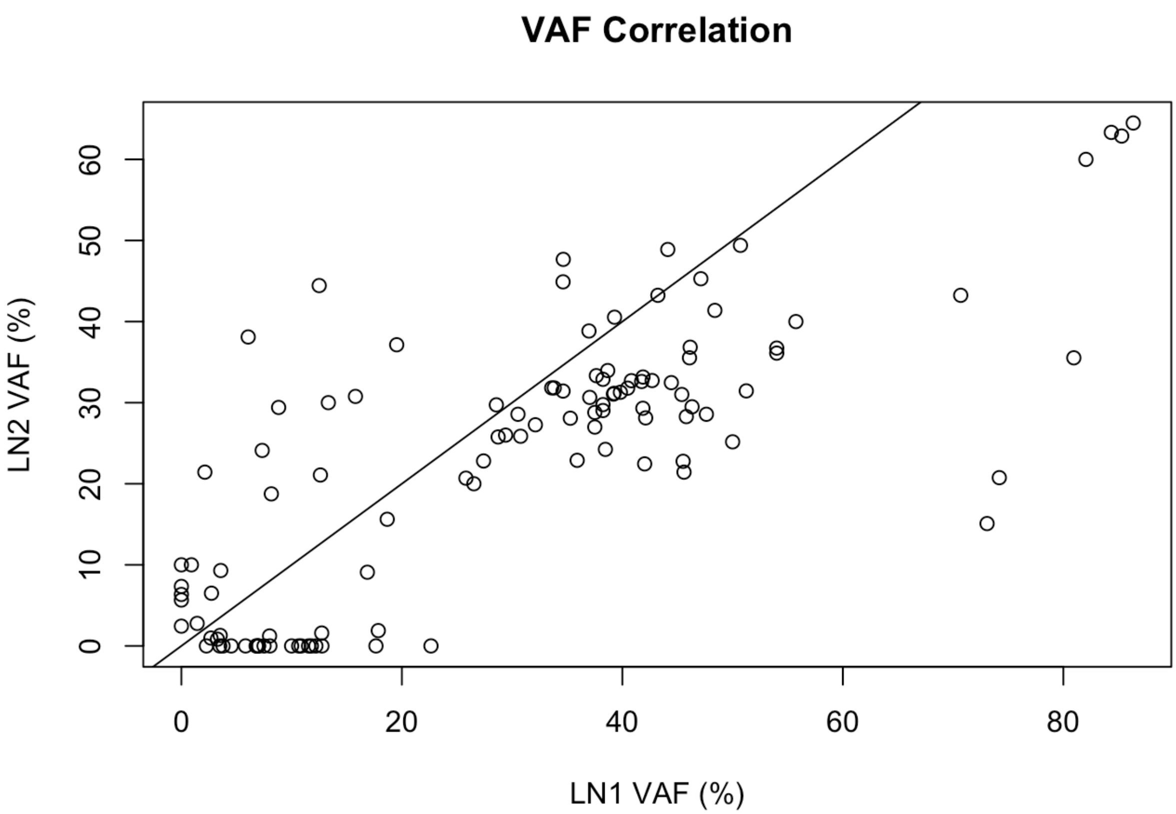


For the paired sample analysis,

(4) Load the "mutations_SP4_LN1_LN2_together.txt" file, and run the mClust paired sample analysis using VAFs of all shared mutations between SP4 LN1 and LN2 samples. First, generate a scatter plot of all VAFs between LN1 and LN2 samples, and briefly comment on the plot on shared mutations between them (**2 marks**).

```
sp4_data <- read.table("mutations_SP4_LN1_LN2_together.txt", header=TRUE, sep="\t")
```

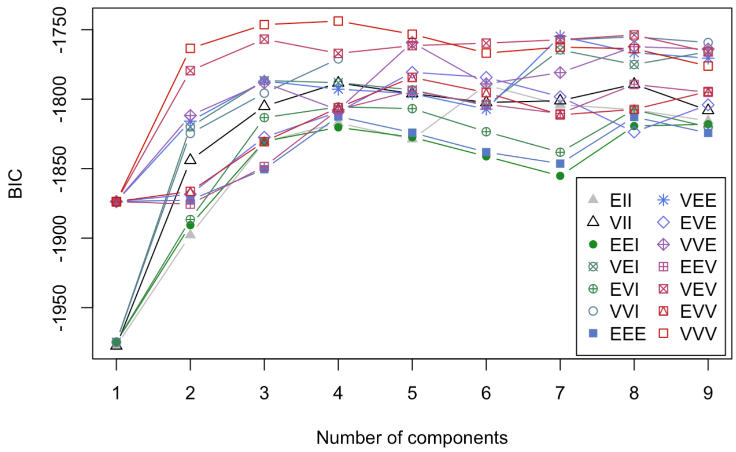
```
sp4_data_vaf <- data.frame(sp4_data$VAF_LN1, sp4_data$VAF_LN2)

plot(sp4_data$VAF_LN1, sp4_data$VAF_LN2,
     xlab="LN1 VAF (%)",
     ylab="LN2 VAF (%)",
     main="VAF Correlation")

#Line (mutations with same VAF in LN1 and LN2)

abline(0,1)
```

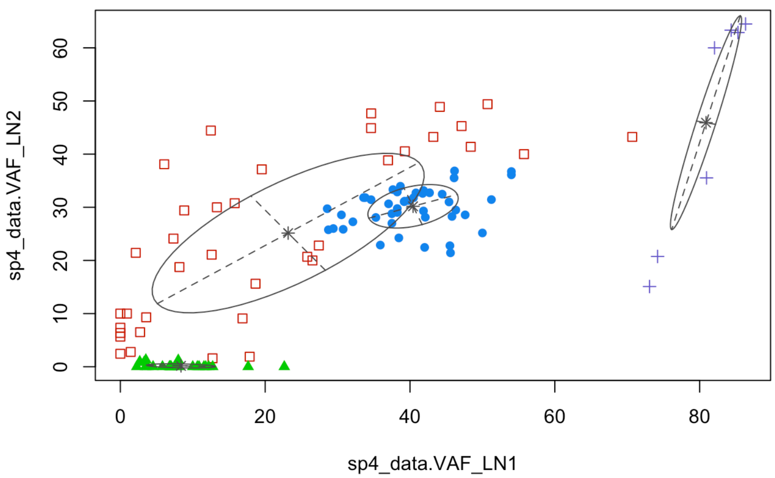


The scatter plot of VAFs between LN1 and LN2 shows that many mutations lie close to the diagonal line, indicating similar VAF values in both samples. These likely represent shared clonal mutations present in both metastatic sites. However, some mutations are positioned away from the diagonal or along the axes (vaf = 0 in one sample and vaf > 0 for the other one). These represent private mutations unique to either LN1 or LN2, suggesting divergent evolution and heterogeneity between the two metastatic sites.

(5) Run mClust analysis based on Gaussian finite mixture model fitted by EM algorithm, followed by dimension reduction clustering and classification analysis. How many clusters can you identify? What is the size (# of mutations) for each cluster? What is the mean VAF value for each cluster in LN1 and LN2 samples, respectively? Can you identify which are major clones (heterozygous or homozygous), which are subclones, and their associated cancer cell fractions (CCF) in LN1 and LN2 samples, respectively? You can summarise all these into a table as shown in the lecture/practical slides (**4 marks**).

```
BIC_sp4 <- mclustBIC(sp4_data_vaf)

plot(BIC_sp4)

mod_sp4 <- Mclust(sp4_data_vaf, x = BIC_sp4)

mod_sp4dr <- MclustDR(mod_sp4)

plot(mod_sp4,what = “classification")

mod_sp4dr$class2mixcomp

cluster_sizes <- table(mod_sp4$classification)

cluster_means <- mod_sp4dr$mu

summary_table_sp4 <- data.frame(

  Cluster = paste("Cluster", 1:4),

  Size = as.vector(cluster_sizes),

  VAF_LN1 = cluster_means[1, ],

  VAF_LN2 = cluster_means[2, ])

trunk_ln1 <- summary_table_sp4$VAF_LN1[summary_table_sp4$Cluster == "Cluster 4"]

trunk_ln2 <- summary_table_sp4$VAF_LN2[summary_table_sp4$Cluster == "Cluster 4"]

summary_table_sp4$ScaledVAF_LN1 <-summary_table_sp4$VAF_LN1 / trunk_ln1

summary_table_sp4$ScaledVAF_LN2 <-summary_table_sp4$VAF_LN2 / trunk_ln2

summary_table_sp4$CCF_LN1 <-summary_table_sp4$ScaledVAF_LN1 * 100

summary_table_sp4$CCF_LN2 <-summary_table_sp4$ScaledVAF_LN2 * 100


summary_table_sp4$Status <- c("Shared Subclone", "Shared Subclone", "Private Subclone
(LN1)", "Trunk")

summary_table_sp4$Zygosity <- c("Heterozygous", "Heterozygous", "Heterozygous",
"Homozygous LN1 - Heterozygous LN2")

print(summary_table_sp4)
```





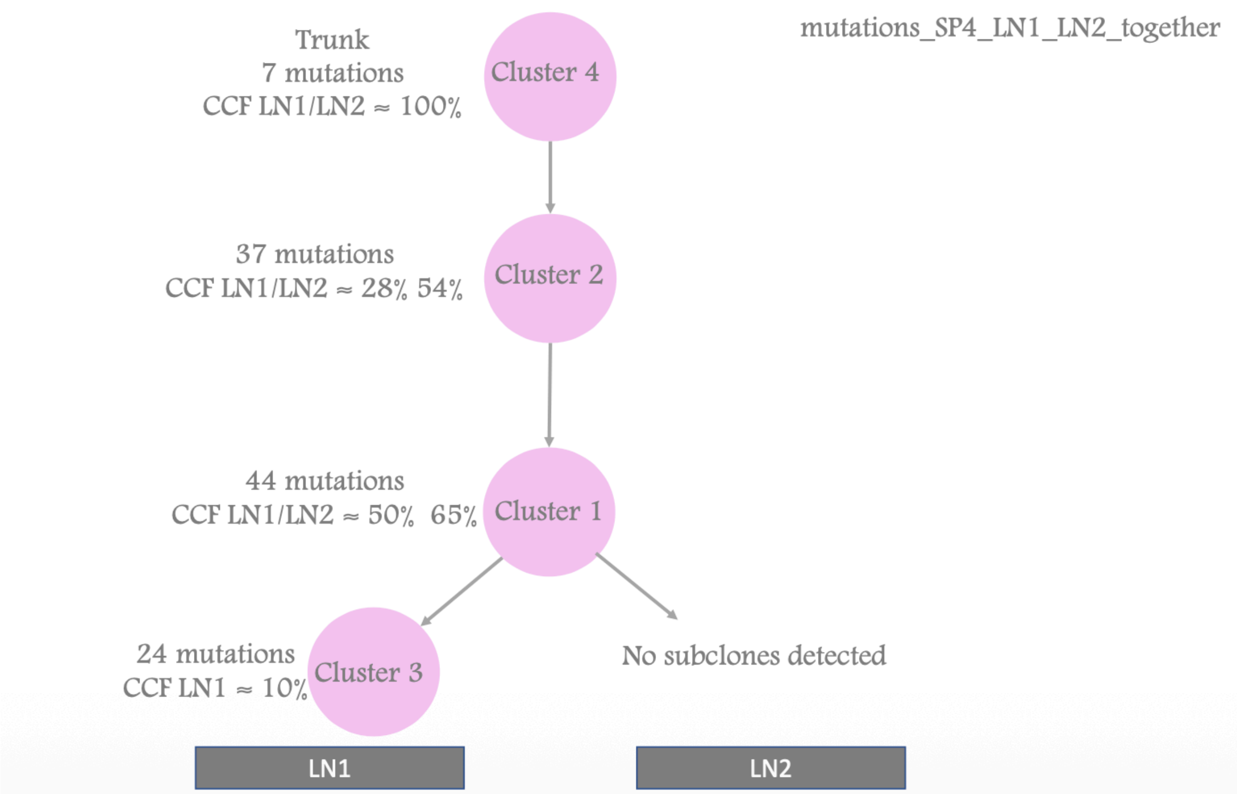

```
##      Cluster Size    VAF_LN1     VAF_LN2 ScaledVAF_LN1 ScaledVAF_LN2    CCF_LN1
## 1 Cluster 1    44 40.420419 30.1858944     0.4997786   0.657426757  49.97786
## 2 Cluster 2    37 23.174340 25.1464052     0.2865393   0.547670360  28.65393
## 3 Cluster 3    24  8.374064  0.1611822     0.1035412   0.003510431  10.35412
## 4 Cluster 4     7 80.876643 45.9152203     1.0000000   1.000000000 100.00000
##        CCF_LN2                 Status                          Zygosity
## 1   65.7426757        Shared Subclone                      Heterozygous
## 2   54.7670360        Shared Subclone                      Heterozygous
## 3    0.3510431 Private Subclone (LN1)                      Heterozygous
## 4  100.0000000                  Trunk Homozygous LN1 - Heterozygous LN2
```

There are four clusters. The number of mutations in each cluster are Cluster 1(44 mutations), Cluster 2(37 mutations), Cluster 3(24 mutations), Cluster 4(7 mutations).

Mean VAF values for LN1 and LN2 were calculated for each cluster and are summarised in the table. Clusters with VAF values around ~0.5 are consistent with heterozygous clonal mutations (major clones), while clusters with lower VAF values represent subclonal populations.

A mutation is considered homozygous when the VAF exceeds 50% (e.g., ~80%), which may indicate that both gene copies are mutated or that loss of heterozygosity (LOH) has occurred. Cancer cell fraction (CCF) for each cluster was estimated based on the VAF values under the diploid assumption and is summarised in the table for LN1 and LN2.

(6) Finally, please draw a tumour evolutionary tree to illustrate the clustering patterns listed above. Within the tree, specifiy clearly the cluster names/numbers and asscoated CCF values in LN1 and LN2 samples, respectively when applicable (**4 marks**).



Cluster	size	Vaf LN1%	Vaf LN2%	CCF LN1 %	CCF LN2 %	Zygosity	Info
1	44	40.420419	30	49.9	65	Heterozygous	Shared Subclone
2	37	23.174340	25	28.65393	54	Heterozygous	Shared Subclone
3	24	8.374064	0.16	10.35412	0.35	Heterozygous	Private Subclone (LN1)
4	7	80.876643	45.9	100	100	Homozygous in LN1/ Heterozygous in LN2	Trunk